Java Application with AWS RDS as PAAS

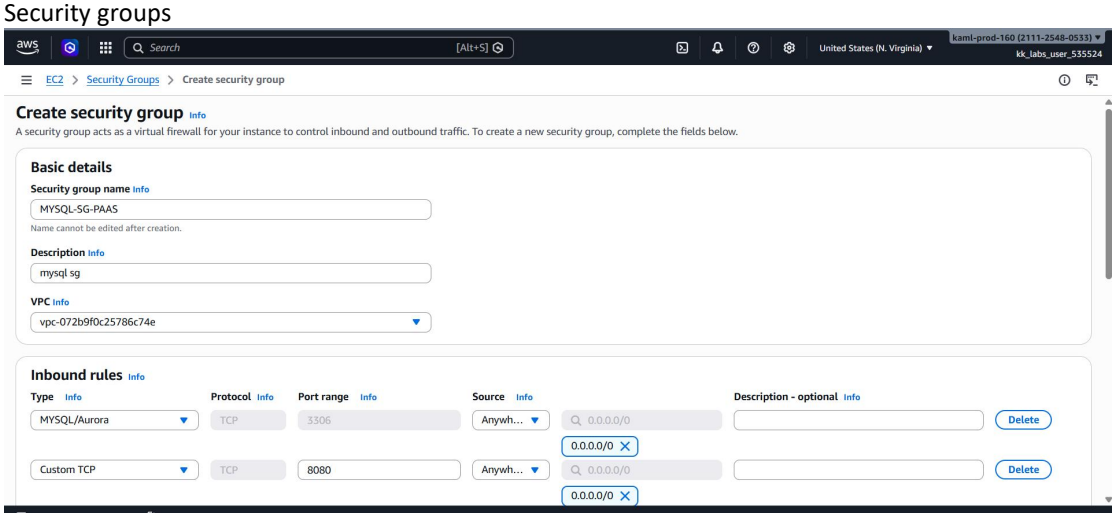
This guide outlines the steps to set up an Ubuntu 24.04 EC2 instance and a MySQL RDS database, ensuring they can communicate through a shared Security Group.

Step 1: Create a Common Security Group

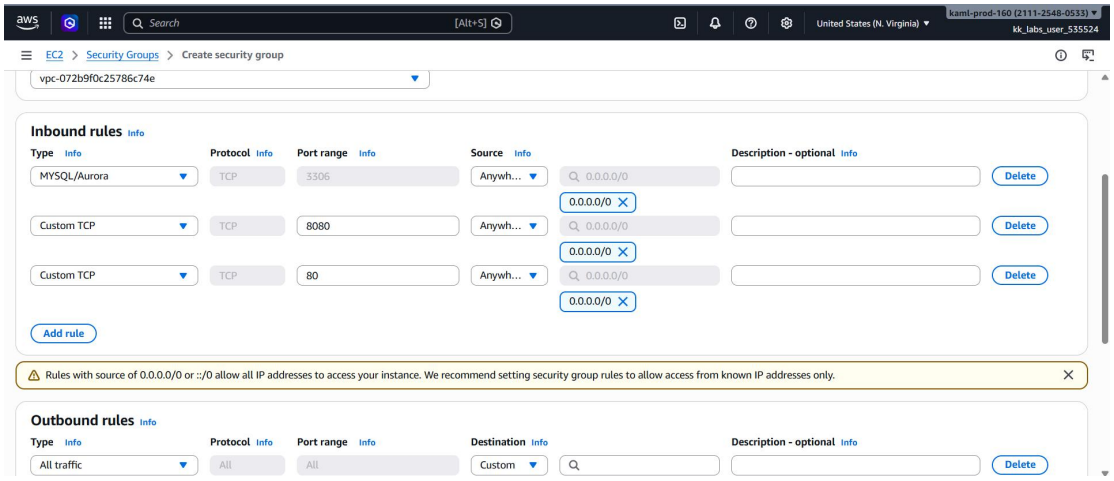
Create this before the RDS and EC2 instances so you can assign it during their creation.

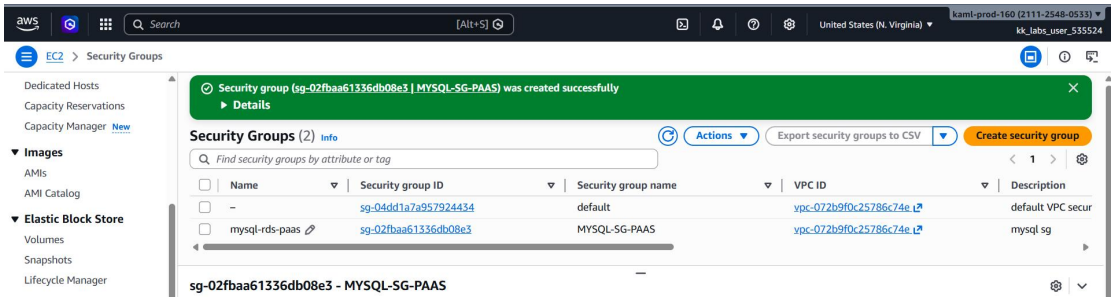
1. Navigate to EC2 Console > Security Groups (left sidebar).
2. Click Create security group.
   Security group name: webappsg
   VPC: Select your default VPC.
3. Inbound rules:
   Rule 1: Type: SSH, Port: 22, Source: AnywhereIPv4 (or My IP for safety).
   Rule 2: Type: HTTP, Port: 80, Source: AnywhereIPv4.
   Rule 3: Type: Custom TCP, Port: 8080 (for Tomcat), Source: AnywhereIPv4.
   Rule 4: Type: MySQL/Aurora, Port: 3306, Source: Select Custom and search for the name of this security group (webappsg) to allow selfreferencing.
4. Click Create security group.

Security groups



Add inbound rules:

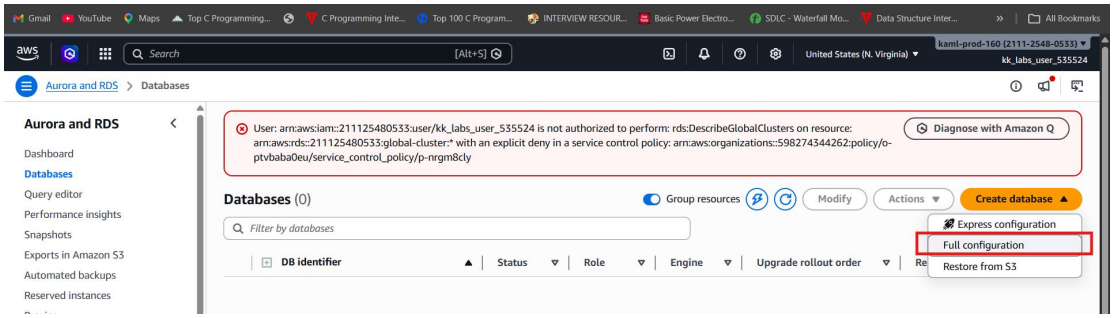


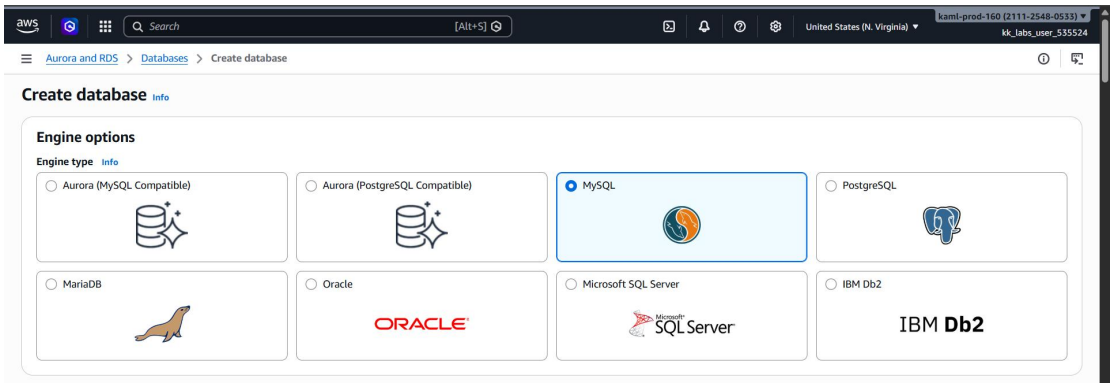


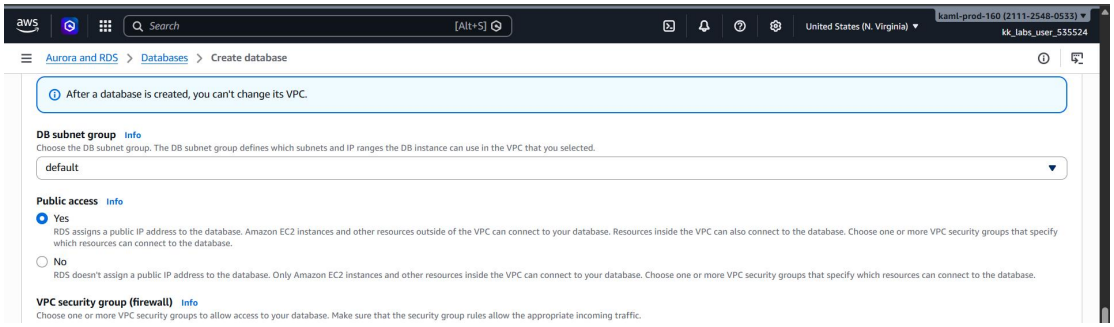
Step 2: Create RDS MySQL Instance

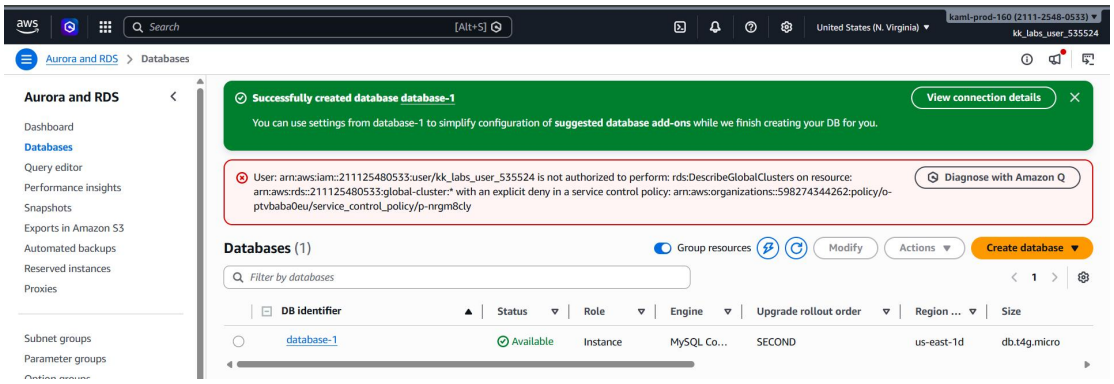
1. Navigate to RDS Console > Databases > Create database.
2. Choose a database creation method: Select Standard create.
3. Engine options: Select MySQL.
4. Templates: Select Free tier.
5. Settings:
   DB instance identifier: mydbinstance.
   Master username: admin.
   Master password: Set a strong password.
6. Instance configuration: Ensure db.t3.micro is selected (default for Free Tier).
7. Connectivity:
   Public access: Select Yes.
   VPC security group: Select Choose existing and pick webappsg.
8. Additional configuration: Under Initial database name, enter login_db.
9. Click Create database. Copy the Endpoint URL once the status is "Available".

Screenshots:

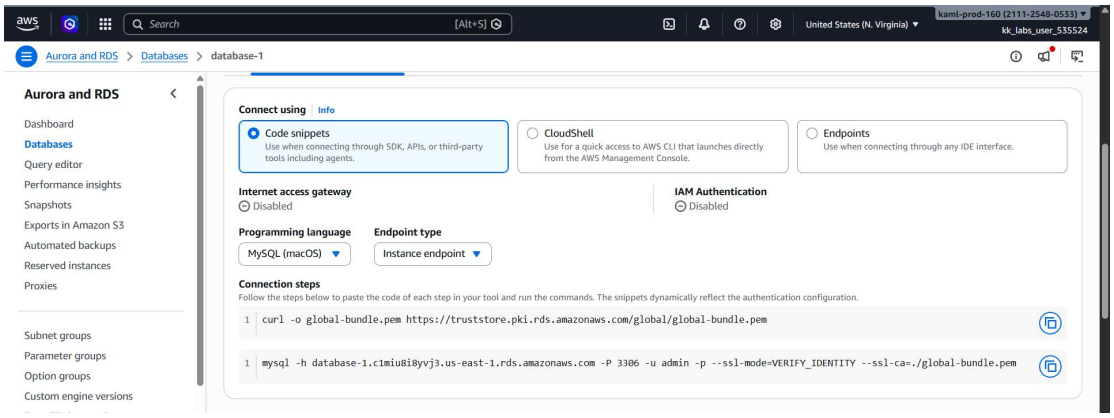








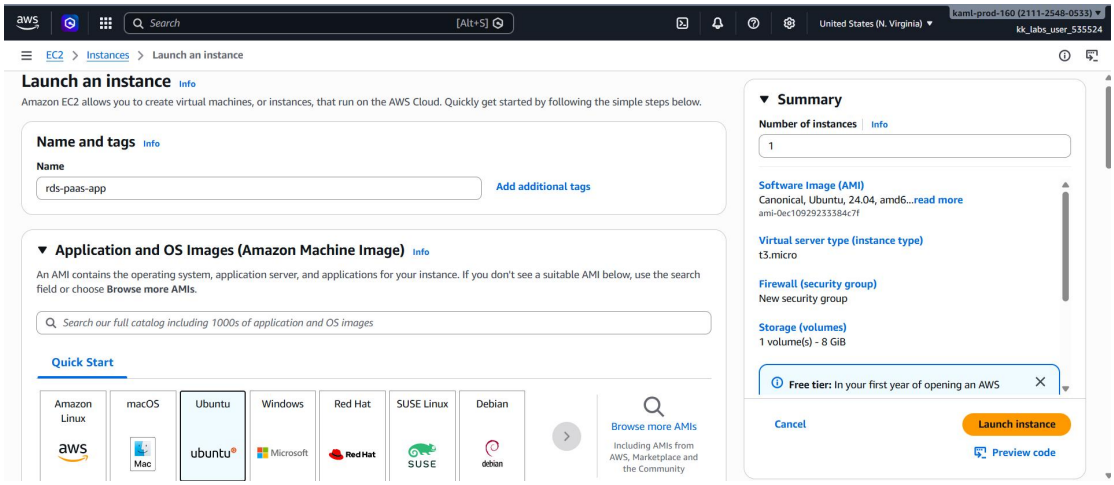
Db string: `database-1.c1miu8i8yvj3.us-east-1.rds.amazonaws.com`

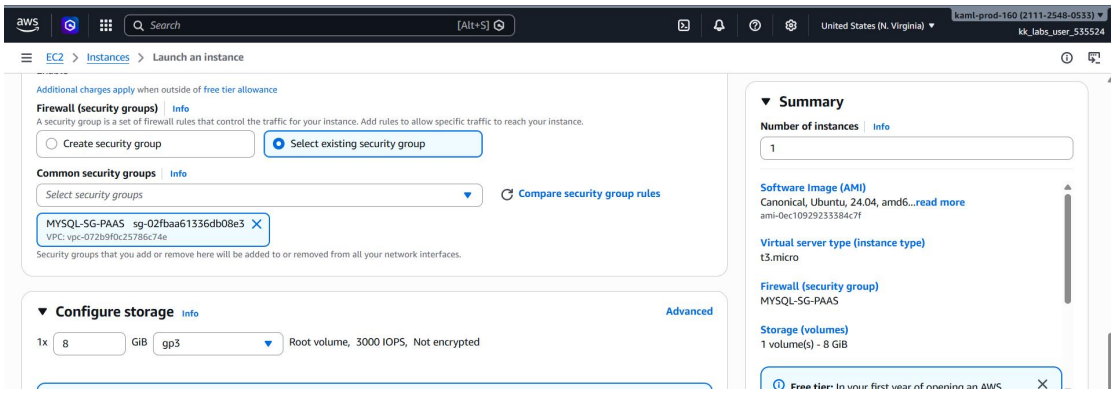


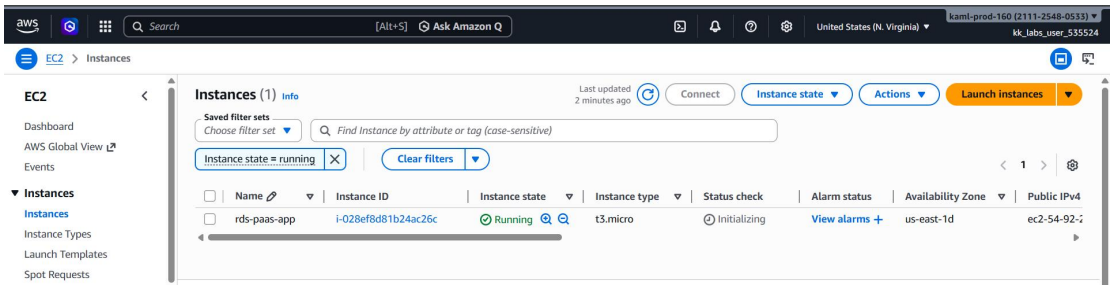
Step 3: Launch Ubuntu EC2 Instance

1. Navigate to EC2 Console > Instances > Launch instances.
2. Application and OS Images (Amazon Machine Image): Select Ubuntu (Ubuntu Server 24.04 LTS).
3. Instance type: Select t3.micro.
4. Key pair: Create or select an existing .pem key for SSH.
5. Network settings:
   Click Edit.
   Security Groups: Select Select existing security group and choose webappsg.
6. Click Launch instance.

Screenshots:







Step 4: Install Software & Build Application

SSH into your Ubuntu server: ssh i yourkey.pem ubuntu@yourec2ip

1. Update and install dependencies:

sudo apt update && sudo apt upgrade y
sudo apt install openjdk17jdk maven mariadbclient git y

2. Install Tomcat 9:

sudo apt install tomcat9 y Start it
sudo systemctl start tomcat9

3. Clone & Build:

git clone https://github.com
cd awsrdsjava

4. Update Database Config:

Update the DBConnection.java or config file in the project with your RDS Endpoint, admin username, and password.

5. Build:

mvn clean package

6. Deploy:

sudo cp target/LoginWebApp.war /var/lib/tomcat9/webapps/

```
[INFO] Webapp assembled in [70 msecs]
[INFO] Building war: /home/ubuntu/aws-rds-java/target/LoginWebApp.war
[INFO] WEB-INF/web.xml already added, skipping
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  11.737 s
[INFO] Finished at: 2026-04-16T09:19:42Z
[INFO] ------------------------------------------------------------------------
ubuntu@app-paas:~/aws-rds-java$ cd
ubuntu@app-paas:~$ mysql -h database-1.c1miu8i8yvj3.us-east-1.rds.amazonaws.com -P 3306 -u admin
ERROR 1045 (28000): Access denied for user 'admin'@'172.31.25.214' (using password: NO)
ubuntu@app-paas:~$ mysql -h database-1.c1miu8i8yvj3.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 40
Server version: 8.4.8 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE jwt;
Query OK, 1 row affected (0.03 sec)

mysql> USE jwt;
```

```
mysql> USE jwt;
Database changed
mysql> CREATE TABLE USER (
    ->   id INT UNSIGNED NOT NULL AUTO_INCREMENT,
    ->   first_name VARCHAR(100) NOT NULL,
    ->   last_name VARCHAR(100) NOT NULL,
    ->   email VARCHAR(150) NOT NULL,
    ->   username VARCHAR(100) NOT NULL,
    ->   password VARCHAR(255) NOT NULL,
    ->   regdate DATE NOT NULL,
    ->   PRIMARY KEY (id),
    ->   UNIQUE KEY unique_username (username),
    ->   UNIQUE KEY unique_email (email)
    -> ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
Query OK, 0 rows affected (0.07 sec)

mysql> show tables;
+---------------+
| Tables_in_jwt |
+---------------+
| USER          |
+---------------+
1 row in set (0.00 sec)
```

```
mysql> use jet;
Database changed
mysql> CREATE TABLE USER (
    ->   id INT UNSIGNED NOT NULL AUTO_INCREMENT,
    ->   first_name VARCHAR(100) NOT NULL,
    ->   last_name VARCHAR(100) NOT NULL,
    ->   email VARCHAR(150) NOT NULL,
    ->   username VARCHAR(100) NOT NULL,
    ->   password VARCHAR(255) NOT NULL,
    ->   regdate DATE NOT NULL,
    ->   PRIMARY KEY (id),
    ->   UNIQUE KEY unique_username (username),
    ->   UNIQUE KEY unique_email (email)
    -> ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
Query OK, 0 rows affected (0.07 sec)

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| jet                |
| jwt                |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.00 sec)

mysql>
```

```
ubuntu@app-paas:~$ wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.110/bin/apache-tomcat-9.0.110.tar.gz
tar -xzf apache-tomcat-9.0.110.tar.gz
cd apache-tomcat-9.0.110/bin
./startup.sh
--2026-04-16 09:26:04--  https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.110/bin/apache-tomcat-9.0.110.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13036068 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.110.tar.gz'

apache-tomcat-9.0.110.tar.gz  100%[===================================================>]  12.43M   113KB/s    in 1m 43s

2026-04-16 09:27:48 (124 KB/s) - 'apache-tomcat-9.0.110.tar.gz' saved [13036068/13036068]

Using CATALINA_BASE:   /home/ubuntu/apache-tomcat-9.0.110
Using CATALINA_HOME:   /home/ubuntu/apache-tomcat-9.0.110
Using CATALINA_TMPDIR: /home/ubuntu/apache-tomcat-9.0.110/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /home/ubuntu/apache-tomcat-9.0.110/bin/bootstrap.jar:/home/ubuntu/apache-tomcat-9.0.110/bin/tomca
t-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

Update Login.jsp

```
  GNU nano 7.2                                                    login.jsp
<%@ page import="java.sql.*"%>
<%
    String userName = request.getParameter("userName");
    String password = request.getParameter("password");
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection("jdbc:mysql://database-1.c1miu8i8yvj3.us-east-1.rds.amazonaws.com:3306/jwt","admin", "admin123");
    Statement st = con.createStatement();
    ResultSet rs;
    rs = st.executeQuery("select * from USER where userName='" +userName + "' and password='" +password + "'");
    if (rs.next()) {
        session.setAttribute("userName", userName);
        response.sendRedirect("success.jsp");
    } else {
        out.println("Invalid password <a href='index.jsp'>try again</a>");
    }
%>
```

Update UserRegistration.jsp:

```
GNU nano 7.2                                                    userRegistration.jsp
<%@ page import="java.sql.*"%>
<%
    String userName = request.getParameter("userName");
    String password = request.getParameter("password");
    String firstName = request.getParameter("firstName");
    String lastName = request.getParameter("lastName");
    String email = request.getParameter("email");
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection("jdbc:mysql://database-1.c1miu8i8yvj3.us-east-1.rds.amazonaws.com:3306/jwt", "admin", "admin123");
    Statement st = con.createStatement();
    int i = st.executeUpdate("insert into USER(first_name, last_name, email, username, password, regdate) values ('" + firstName + "','"
    + lastName + "','" + email + "','" + userName + "','" + password + "', CURDATE())");
    if (i > 0) {
        response.sendRedirect("welcome.jsp");

    } else {
        response.sendRedirect("index.jsp");
    }
%>
```

Add user

Add user in tomcat-users.xml

```
ubuntu@app-paas:~/apache-tomcat-9.0.110/conf$ ls
Catalina         catalina.properties  jaspic-providers.xml  logging.properties  tomcat-users.xml  web.xml
catalina.policy  context.xml          jaspic-providers.xsd  server.xml          tomcat-users.xsd
ubuntu@app-paas:~/apache-tomcat-9.0.110/conf$ nano tomcat-users.xml
ubuntu@app-paas:~/apache-tomcat-9.0.110/conf$
<!--
  <user username="admin" password="<must-be-changed>" roles="manager-gui"/>
  <user username="robot" password="<must-be-changed>" roles="manager-script"/>
-->
<!--
  The sample user and role entries below are intended for use with the
  examples web application. They are wrapped in a comment and thus are ignored
  when reading this file. If you wish to configure these users for use with the
  examples web application, do not forget to remove the <!.. ..> that surrounds
  them. You will also need to set the passwords to something appropriate.
-->
<!--
  <role rolename="tomcat"/>
  <role rolename="role1"/>
  <user username="tomcat" password="<must-be-changed>" roles="tomcat"/>
  <user username="both" password="<must-be-changed>" roles="tomcat,role1"/>
  <user username="role1" password="<must-be-changed>" roles="role1"/>
-->
</tomcat-users>
<role rolename="manager-gui"/>
<role rolename="manager-script"/>
<user username="admin" password="admin123" roles="manager-gui,manager-script"/>
```

Edit Meta-Inf

```
ubuntu@app-paas:~/apache-tomcat-9.0.110/webapps$ ls
LoginWebApp  LoginWebApp.war  ROOT  docs  examples  host-manager  manager
ubuntu@app-paas:~/apache-tomcat-9.0.110/webapps$ cd manager/
ubuntu@app-paas:~/apache-tomcat-9.0.110/webapps/manager$ ls
META-INF  WEB-INF  css  images  index.jsp  status.xsd  xform.xsl
ubuntu@app-paas:~/apache-tomcat-9.0.110/webapps/manager$ cd META-INF/
ubuntu@app-paas:~/apache-tomcat-9.0.110/webapps/manager/META-INF$ LS
LS: command not found
ubuntu@app-paas:~/apache-tomcat-9.0.110/webapps/manager/META-INF$ ls
context.xml
ubuntu@app-paas:~/apache-tomcat-9.0.110/webapps/manager/META-INF$
```

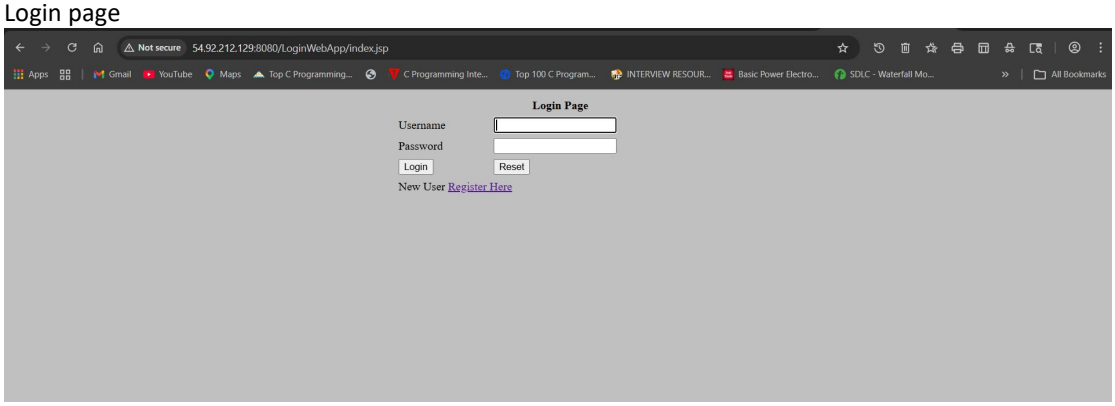
Comment out

```
GNU nano 7.2                                                    context.xml
<?xml version="1.0" encoding="UTF-8"?>
<!--
  Licensed to the Apache Software Foundation (ASF) under one or more
  contributor license agreements.  See the NOTICE file distributed with
  this work for additional information regarding copyright ownership.
  The ASF licenses this file to You under the Apache License, Version 2.0
  (the "License"); you may not use this file except in compliance with
  the License.  You may obtain a copy of the License at

      http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License.
-->
<!--
<Context antiResourceLocking="false" privileged="true" >
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
                   sameSiteCookies="strict" />
  <Valve className="org.apache.catalina.valves.RemoteAddrValve"
         allow="127\.\d+\.\d+\.\d+|::1|0:0:0:0:0:0:0:1" />
  <Manager sessionAttributeValueClassNameFilter="java\.lang\.(?:Boolean|Integer|Long|Number|String)|org\.apache\.catalina\.filters\.CsrfPreventionFilter\$LS
</Context>
-->
```

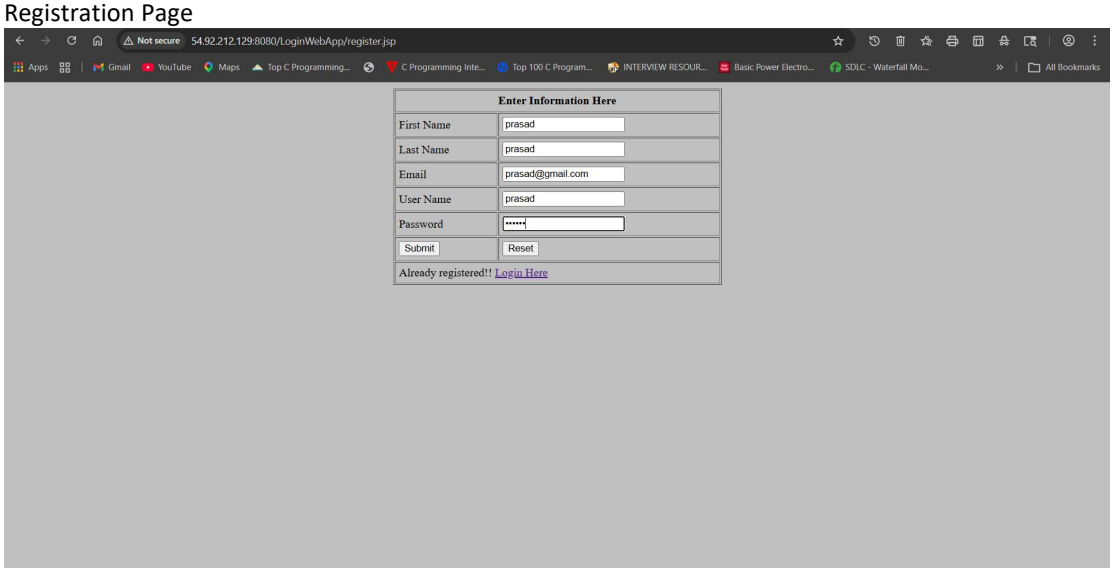
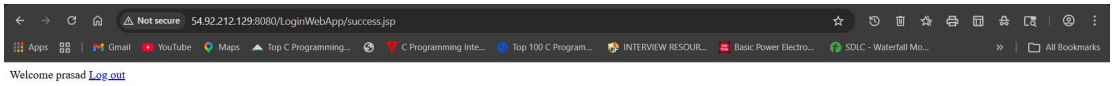
Validation:
Step 5: Test Connection

Use the installed client to verify connectivity to RDS from the EC2:

http://54.92.212.129:8080/LoginWebApp/register.jsp
Login page



Registration Page



User login